

AMANDEEP SINGH

Software Developer | Node.js & AI Integration

Ferozpur, Punjab | +91 94648 20510 | amandeep2975@gmail.com
GitHub: <https://github.com/Aman2975>
Portfolio: <https://iaman.space/>
LinkedIn: <http://www.linkedin.com/in/amandeep2975>

SUMMARY

MCA fresher specializing in backend development with hands-on experience building production-grade systems. Built ShodhKartaChecker (AI-powered plagiarism detection engine) using Node.js, FastAPI, Redis task queues, and Groq AI, achieving 40% faster API response times.

Skilled in microservices architecture, JWT authentication, RBAC, and REST API design. Looking to contribute to a backend-focused team and solve real-world engineering problems.

SKILLS

- Languages: JavaScript (Node.js), Python, TypeScript
- Backend: REST APIs, Microservices, JWT Authentication, RBAC, Swagger, Async Processing, Caching Redis Queues, AI API Integration
- Frameworks: Express.js, React.js, NestJS
- Databases: MySQL, MongoDB, Redis, PostgreSQL
- ORMs: Prisma, Mongoose, TypeORM
- DevOps: Docker, Nginx, PM2, Linux, Git, GitHub, CI/CD
- Testing: Jest, Supertest, Postman
- Cloud & Storage: Cloudinary, MongoDB Atlas
- Deployment: VPS, Vercel, Render
- Concepts: System Design, Data Structures, DB Design, Error Handling, MVC Architecture, Backend Optimization, Asynchronous Processing, Concurrency

CERTIFICATION AND TRAINING

Full Stack Web Development Training – Department of Computer Science Punjabi University, Patiala

(MCA Industrial Training Program)

Duration: 1 Jan 2026 – 5 May 2026

- Learned and worked JavaScript, React.js, Node.js, Express.js, MongoDB, Python and Redis
- Developed backend applications and AI-integrated projects during training
- Built an AI-Powered Plagiarism Detection System with Redis-based task processing and Groq AI API integration

PROJECTS

ShodhKartaChecker (AI-Powered Plagiarism Detection System) | [GitHub](#) | [Live Demo](#)

- Built a distributed plagiarism detection platform integrating **2 backend services** (Node.js and FastAPI).

- Implemented **Redis-based asynchronous task queues**, reducing API response time by **40%**.
- Architected a microservices-based workflow for **document upload, preprocessing, AI analysis, and report generation**.
- Integrated **Groq AI APIs** for plagiarism explanations.
- Deployed the application using **Docker, Nginx, and PM2** on a VPS.

Schedula (Doctor appointment Booking System) [GitHub](#) | [Live Demo](#)

- Created **20+ REST API endpoints** using NestJS and TypeScript.
- Designed a normalized PostgreSQL schema with **4+ relational tables** for users, doctors, appointments, and schedules.
- Implemented JWT authentication with **2 user roles** (Doctor and Patient).
- Added pagination and filtering to improve doctor search performance.
- Documented **REST APIs** using Swagger UI.

PostGrid (Blog Platform) [GitHub](#) | [Live Demo](#)

- Developed **15+ REST API endpoints** supporting CRUD operations, authentication, and profile management.
- Integrated Cloudinary for cloud media storage and Multer for multipart file uploads.
- Implemented a PostgreSQL database with **3+ relational models** using Prisma ORM.
- Containerized the application using Docker for consistent deployment.
- Implemented JWT authentication for secure user access.

EDUCATION

Master of Computer Applications (MCA)
Punjabi University – Patiala, India
Completed: May 2026
CGPA:8.1